

Raihan Ghifari Winata

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PROFILE

Machine Learning & AI Engineer specializing in Deep Reinforcement Learning, Generative AI (LLMs/VLMs), and end-to-end MLOps. Proven track record of accelerating PyTorch training pipelines by 10x on AWS cloud infrastructure and engineering safety-constrained models that outperform baseline benchmarks by 14%. Skilled in building production-grade AI applications using FastAPI, LangChain/LangGraph, and Docker. Certified AWS AI Practitioner seeking to deploy safe, scalable, and interpretable AI solutions globally.

EDUCATION

Bachelor of Computer Science, Faculty of Information Technology 09/2022 – 04/2026 | Jakarta
Universitas Yarsi
CGPA: 3.92 of 4.00
Thesis: *Development of a Soft Actor-Critic Model Based on Ensemble Q-Learning for Sepsis Treatment Recommendation with Continuous Action Representation*
Relevant Courses: *Deep Learning & Reinforcement Learning, Artificial Intelligence, Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems, Web Programming, Mobile Development, Calculus, Probability and Statistics, Data Analytics.*

PROFESSIONAL EXPERIENCE

Artificial Intelligence Researcher 08/2025 – 01/2026
Lab AI Universitas Yarsi

- Engineered a rigorous clinical data pipeline that filtered an initial cohort of 94,458 raw ICU stays down to a high-quality dataset of 35,608 episodes by applying Sepsis-3 cohort and handling missing values.
- Designed a denoising Autoencoder to compress 37 high-noise clinical features into a robust 24-dimensional latent space, significantly reducing data dimensionality and improving model convergence.
- Developed an end-to-end Reinforcement Learning architecture that processed the refined dataset (stratified split into 28,486 training and 7,122 validation/testing episodes) to optimize continuous dosage recommendations.
- Achieved a 75.31% estimated survival rate and an expected return of 17.24 outperforming the historical clinician baseline of 73.55% by implementing an ensemble of 5 SAC agents with median action aggregation.
- Enforced Lagrangian Safety Constraints dynamically to ensure policy outputs remained strictly within safe physiological boundaries for Mean Arterial Pressure (MAP) and fluid balance.
- Validated algorithmic reliability using a Calibrated Doubly Robust estimator, achieving a stable Effective Sample Size (ESS) of 1374.

Artificial Intelligence Intern 02/2025 – 07/2025
Lab E-Health Universitas Yarsi

- Architected a temporal data extraction pipeline from the MIMIC-III database, processing 61,532 raw ICU admissions into a refined Sepsis-3 cohort of 20,913 clean clinical trajectories.
- Aggregated 37 complex clinical variables (vital signs, lab results, and demographic details) into precise 4-hour temporal time steps to capture dynamic patient responses.
- Built an Ensemble Weighted Dueling Double Deep Q-Network (EWD3QN) leveraging the preprocessed dataset to evaluate 25 discrete clinical action combinations.
- Achieved a 92.4% estimated survival rate, significantly surpassing the historical clinician baseline of 83.26%.
- Generated a cumulative expected return of 23.6 compared to the clinician baseline of 14.11.
- Accelerated model training time by reducing epoch duration from 20 minutes to approximately 2 minutes using optimized GPU parallel processing.
- Utilized Smooth L1 Loss (Huber Loss) to minimize temporal difference errors and improve model resilience against clinical data outliers.

Assistant Lecturer / Lab Instructor 09/2023 – 04/2026
Universitas Yarsi

- Mentored 100+ students across 5 core Computer Science courses (Specialization AI Track, Artificial Intelligence, Data Structures, OOP, Algorithm Fundamentals).
- Led 14 weekly lab sessions per semester, translating complex AI and Deep Learning architectures into practical Python/Java programming assignments.
- Conducted rigorous technical code reviews and provided model performance feedback, which contributed to a 70% increase in average student project scores.

AWARDS

PRAGMA Collaborative Hackathon Award 01/2026
Thammasat University
Winner Teamwork Award at a 15-university Asia-Pacific hackathon involving UCSD and Osaka University

Samsung Innovation Campus 09/2024
Skilvul
Ranked as Semi-Finalist in a national Samsung-sponsored AI competition among 5,000+ participants.

ACADEMIC & INDEPENDENT PROJECTS

Transformer-Based Clinical Decision Support System
(PRAGMA Hackathon Winner)

- Co-developed a Decision Transformer pipeline with Thammasat University students at the PRAGMA Hackathon to optimize ICU treatment strategies.
- Engineered a sequence-to-sequence model that predicts continuous drug dosages based on historical patient states and target health outcomes.
- Processed multi-step physiological time-series data to capture long-term trends in patient recovery and disease progression.
- Optimized training stability using cross-entropy and policy loss functions for accurate dosage generation across high-dimensional state spaces.
- Validated treatment policies against baseline models using off-policy evaluation (OPE) metrics on retrospective clinical trajectories.

- Collaborated with international researchers to utilize distributed computing infrastructure, significantly accelerating model training.

Automated Industrial Defect Triage System (Vision-Language Models & Computer Vision)

- Architected an automated vision-language defect triage system combining Vision-Language Models (VLMs) and Computer Vision to classify and inspect structural defects from image feeds.
- Engineered a hybrid inference pipeline leveraging zero-shot VLM prompting alongside downstream CV object detection to extract spatial coordinates and classify multi-class defect severity.
- Developed a custom data preprocessing and augmentation workflow to normalize variable-lighting visual inputs and balance low-frequency defect class distributions.
- Optimized model inference latency and throughput by implementing asynchronous batching and quantization, enabling real-time edge processing for automated quality inspection.
- Evaluated system precision and recall across edge-case defect scenarios, achieving robust classification accuracy over traditional rule-based computer vision baselines.
- Containerized the end-to-end vision pipeline using Docker and deployed API endpoints for seamless integration into automated industrial monitoring workflows.

ORGANISATIONS

Faculty of Information Technology Student Senate of YARSI University
Head of Communication Department

08/2023 – 12/2024

- Led a creative team of 10 members in managing publication, digital archives, and marketing collateral for major campus events serving over 60 attendees.

Directorate of Distance Learning of YARSI University
Lead Student Mentor (University Orientation)

09/2023 – 02/2024

- Guided 12 incoming freshmen through academic orientation, providing mentorship on program structures and goal-setting.

Pacific Rim Application and Grid Middleware Assembly (PRAGMA)
Technical Committee

06/2023 – 06/2023

- Served on technical committee for an international IT collaboration event involving 15 Asia-Pacific universities (UCSD, Osaka University, and others). Acted as liaison between international researchers and the organizing team.

CERTIFICATES

CITI Data or Specimens Only Research	CITI Conflicts of Interest	AWS Certification AI Practitioner Score: 826	TOEIC Score: 605
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SKILLS

AI & Machine Learning Engineering

- Generative AI & LLM Frameworks: LLMs, LangChain, LangGraph, Langflow, Prompt Engineering
- Core AI & Architectures: Deep Reinforcement Learning (DQN, SAC, TD3), Computer Vision (YOLO, CNN, VLM), Offline Evaluation (OPE, WIS, DR)
- ML Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn

Data Analytics & Visualization

- Data Processing & Analysis: Pandas, NumPy, Feature Engineering, Exploratory Data Analysis (EDA)
- Business Intelligence & Visualization: Power BI, Tableau, Matplotlib, Seaborn

Programming & Development

- Languages: Python, SQL, JavaScript, HTML/CSS
- Frontend Development: Next.js, React.js, Vue.js, Tailwind CSS
- Backend & API Development: FastAPI, Flask REST API, RESTful APIs
- Database & Cloud Infrastructure: PostgreSQL, NoSQL, MongoDB Atlas, AWS, Docker, GPU Parallel Processing

Collaboration & Productivity

- Developer & Management Tools: Git, GitHub, Jira, Postman, Linux/Bash
- Documentation & Communication: Meeting tools (Zoom, Gmeet, Ms. Teams) , Canva (Visual Reports), Google Workspace, Microsoft Office

COURSES

AWS Machine Learning Foundations
AWS

09/2025

Algorithm & Data Structures with Python
Skilvul

03/2024

Advanced Python
Skilvul

03/2024

LANGUAGES

- English
- Indonesia